The Anatomy of Agentic Coding

How an agent structures its loop — and where the safety boundaries live.

THE LOOP

The Prompt Loop

Read → Plan → Write → Verify — then repeat using execution feedback.

- Read: the agent loads repository context
- Plan: it chooses the next bounded action
- Write: it makes multi-file edits
- Verify: it runs checks and reads the result
- Each retry uses what the last attempt learned.

THE TOOLS

Tools connect models to the world

Files · commands · search · tests — the interface shown is illustrative.

- A tool is a named, permissioned capability the model can call.
- The shown names (read / edit / run / search) are illustrative, not a specific product’s API.
- What matters is the contract: input in, result back, model decides the next step.

THE NUMBER

One benchmark, read carefully

The 2023 Claude 2 + BM25 result on full SWE-bench is 1.96%.

- Source: SWE-bench paper, arXiv:2310.06770v1, Table 5.
- Setup: Claude 2 with BM25 retrieval, full SWE-bench.
- What it IS: a historical baseline.
- What it is NOT: SWE-bench Lite, a current leaderboard, or loop uplift.

Retrying with feedback is qualitative, not a measured gain.

THE BOUNDARY

Human review matches risk

Autonomy needs boundaries — calibrated to consequence.

- Match review to the action’s consequence, reversibility, and scope.
- A risky, irreversible, broad action needs a human; a safe, reversible, narrow one may not.
- This is risk-based review, not “approve everything” or “approve nothing.”